

4.3 CONTINUOUS-TIME SYSTEMS

In this section, we will look at how the mean and covariance of the state of a continuous-time linear system propagate. Consider the continuous-time system

$$\dot{x} = Ax + Bu + w \quad (4.34)$$

where $u(t)$ is a known control input and $w(t)$ is zero-mean white noise with a covariance of

$$E[w(t)w^T(\tau)] = Q_c\delta(t - \tau) \quad (4.35)$$

By taking the mean of Equation (4.34), we can obtain the following equation for the derivative of the mean of the state:

$$\dot{\bar{x}} = A\bar{x} + Bu \quad (4.36)$$

This equation shows how the mean of the state propagates with time. The linear equation that describes the propagation of the mean looks very much like the original state equation, Equation (4.34). We can also obtain Equation (4.36) by using the equation that describes the mean of a sampled-data system and taking the limit as $\Delta t = t_k - t_{k-1}$ goes to zero. Taking the mean of Equation (4.18) gives

$$\bar{x}_k = e^{A\Delta t}\bar{x}_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}B(\tau)u(\tau)d\tau \quad (4.37)$$

The state transition matrix can be written as

$$\begin{aligned} F &= e^{A\Delta t} \\ &= I + A\Delta t + \frac{(A\Delta t)^2}{2!} + \dots \end{aligned} \quad (4.38)$$

For small values of Δt , this can be approximated as

$$F \approx I + A\Delta t \quad (4.39)$$

With this substitution Equation (4.37) becomes

$$\bar{x}_k = (I + A\Delta t)\bar{x}_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}B(\tau)u(\tau)d\tau \quad (4.40)$$

Subtracting $\bar{x}_{k-1}$ from both sides and dividing by Δt gives

$$\frac{\bar{x}_k - \bar{x}_{k-1}}{\Delta t} = A\bar{x}_{k-1} + \frac{1}{\Delta t} \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}B(\tau)u(\tau)d\tau \quad (4.41)$$

Taking some limits as Δt goes to zero gives the following:

$$\begin{aligned} \lim_{\Delta t \rightarrow 0} \frac{\bar{x}_k - \bar{x}_{k-1}}{\Delta t} &= \dot{\bar{x}} \\ \lim_{\Delta t \rightarrow 0} e^{A(t_k-\tau)} &= I \text{ for } \tau \in [t_{k-1}, t_k] \end{aligned} \quad (4.42)$$

Making these substitutions in (4.41) gives

$$\dot{\bar{x}} = A\bar{x} + Bu \quad (4.43)$$

which is the same equation as the one we derived earlier in Equation (4.36) by a more direct method. Although the limiting argument that we used here was not necessary because we already had the mean equation in Equation (4.36), this method shows us how we can use limiting arguments (in general) to obtain continuous-time formulas.

Next we will use a limiting argument to derive the covariance of the state of a continuous-time system. Recall the equation for the covariance of a sampled data system from Equation (4.26):

$$P_k = F_{k-1}P_{k-1}F_{k-1}^T + Q_{k-1} \quad (4.44)$$

For small Δt we again approximate F_{k-1} as shown in Equation (4.39) and substitute into the above equation to obtain

$$\begin{aligned} P_k &\approx (I + A\Delta t)P_{k-1}(I + A\Delta t)^T + Q_{k-1} \\ &= P_{k-1} + AP_{k-1}\Delta t + P_{k-1}A^T\Delta t + AP_{k-1}A^T(\Delta t)^2 + Q_{k-1} \end{aligned} \quad (4.45)$$

Subtracting P_{k-1} from both sides and dividing by Δt gives

$$\frac{P_k - P_{k-1}}{\Delta t} = AP_{k-1} + P_{k-1}A^T + AP_{k-1}A^T\Delta t + \frac{Q_{k-1}}{\Delta t} \quad (4.46)$$

Recall from Equation (4.28) that for small Δt

$$Q_{k-1} \approx Q_c(t_k)\Delta t \quad (4.47)$$

This can be written as

$$\frac{Q_{k-1}}{\Delta t} \approx Q_c(t_k) \quad (4.48)$$

Therefore, taking the limit of Equation (4.46) as Δt goes to zero gives

$$\dot{P} = AP + PA^T + Q_c \quad (4.49)$$

This continuous-time Lyapunov equation, also sometimes called a Sylvester equation, gives us the equation for how the covariance of the state of a continuous-time system propagates with time.

It is interesting to consider the conditions under which the continuous-time Lyapunov equation has a steady-state solution. That is, suppose that $A(t) = A$ is a constant, and $Q_c(t) = Q_c$ is a constant. Then we have the following theorem, whose proof can be found in [Kai00, Appendix D].

Theorem 22 *Consider the equation $AP + PA^T + Q_c = 0$ where A and Q_c are real matrices. Denote by $\lambda_i(A)$ the eigenvalues of the A matrix.*

1. *A unique solution P exists if and only if $\lambda_i(A) + \lambda_j(A) \neq 0$ for all i, j . This unique solution is symmetric.*
2. *Note that the above condition includes the case of stable A , because if A is stable then all of its eigenvalues have real parts less than 0, so $\lambda_i(A) + \lambda_j(A) \neq 0$ for all i, j . Therefore, we see that if A is stable then the continuous-time Lyapunov equation has a solution P that is unique and symmetric. In this case, the solution can be written as*

$$P = \int_0^\infty e^{A^T\tau} Q_c e^{A\tau} d\tau \quad (4.50)$$

3. If A is stable and Q_c is positive (semi)definite, then the unique solution P is symmetric and positive (semi)definite.
4. If A is stable, Q_c is positive semidefinite, and $[A, (Q_c^{1/2})^T]$ is controllable, then P is unique, symmetric, and positive definite. Note that $Q_c^{1/2}$, the square root of Q_c , is defined here as any matrix such that $Q_c^{1/2}(Q_c^{1/2})^T = Q_c$.

■ EXAMPLE 4.3

Suppose we have the first-order, continuous-time dynamic system given by Equation (4.29):

$$\begin{aligned}\dot{x} &= fx + w \\ E[w(t)w(t+\tau)] &= q_c\delta(\tau)\end{aligned}\tag{4.51}$$

where $w(t)$ is zero-mean noise. The equation for the continuous-time propagation of the mean of the state is obtained from Equation (4.36):

$$\dot{\bar{x}} = f\bar{x}\tag{4.52}$$

Solving this equation for $\bar{x}(t)$ gives

$$\bar{x}(t) = \exp(ft)\bar{x}(0)\tag{4.53}$$

We see that the mean will increase without bound if $f > 0$ (i.e., if the system is unstable), but the mean will asymptotically tend to zero if $f < 0$ (i.e., if the system is stable). The equation for the continuous-time propagation of the covariance of the state is obtained from Equation (4.49):

$$\dot{P} = 2fP + q_c\tag{4.54}$$

Solving this equation for $P(t)$ gives

$$P(t) = \left(P(0) + \frac{q_c}{2f}\right)\exp(2ft) - \frac{q_c}{2f}\tag{4.55}$$

We see that the covariance will increase without bound if $f > 0$ (i.e., if the system is unstable), but the covariance will asymptotically tend to $-q_c/2f$ if $f < 0$ (i.e., if the system is stable). Compare these results with Example 4.2.

The steady-state value of P can also be computed using Equation (4.50). If we substitute f for A and q_c for Q_c in Equation (4.50), we obtain

$$\begin{aligned}P &= \int_0^\infty e^{2f\tau} q_c d\tau \\ &= \frac{q_c}{2f} e^{2f\tau} \Big|_0^\infty\end{aligned}\tag{4.56}$$

The integral converges for $f < 0$ (i.e., if the system is stable), in which case $P = -q_c/2f$.

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